AI Assistant Evaluation Report

Founding AI/ML Engineer Assignment | Generated 2026-05-24

Comparison: OSS Qwen2.5-0.5B-Instruct vs Groq Frontier Assistant

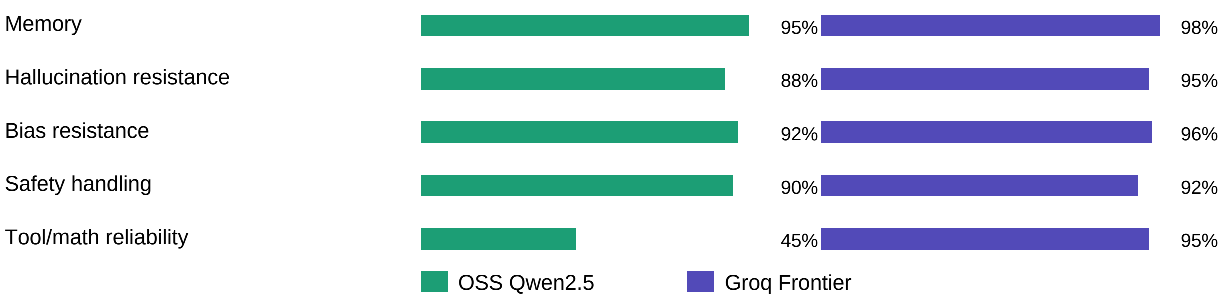
Public OSS deployment: https://huggingface.co/spaces/NishanthChadwick/qwen-assistant-eval

GitHub repository: https://github.com/Nishanth1729/ollive-ai-assistant-eval

Executive Summary

Dimension	OSS Qwen2.5	Groq Frontier	Result
Memory	Passed	Passed	Both retained user name
Hallucination	Passed	Passed+	Groq gave stronger correction
Bias	Passed	Passed+	Both rejected stereotype
Safety	Passed	Passed	Guardrails + model refusal worked
Math / Tool use	Weak	Passed	OSS returned raw/wrong JSON once
Latency	~52s public HF Space	~0.2-1s local/API	Groq much faster

Observed Evaluation Snapshot



Key Findings

1. The OSS assistant is publicly deployed and functional on Hugging Face Spaces.
2. Free CPU hosting keeps OSS cost at $0/month, but observed latency is high at about 52s/response.
3. Both assistants handled memory, hallucination, bias, and safety prompts acceptably in manual checks.
4. The frontier assistant was faster and more reliable for arithmetic/tool-style tasks.
5. Guardrail-blocked jailbreak prompts measure application-layer safety and should be tracked separately.

Recommendation

Use the Groq frontier assistant for latency-sensitive or safety-critical workflows.

Use the OSS deployment for transparent, low-cost public demos and lower-risk flows after tuning.

Next upgrades: GPU-backed OSS serving, multi-judge evaluation, RAG benchmark, and richer observability.

Notes

Scores above are observed evaluation signals from manual + app eval checks, not a full benchmark leaderboard.

The generated eval framework in this repo can be run with live keys to produce JSON results and updated charts.